

Linear Algebra

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1 Linear Algebra Through Matrices

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The prerequisites for this course include a sound knowledge of matrices up to the level of high school. This also entails some basic knowledge of linear algebra including solving linear systems of equations by hand, possibly up to 3 variables.

Knowledge of set theory is crucial, too. We also make the remark that for the rest of this course, all vectors shall be treated as column vectors and consequently multiplication of matrices will be done from the left side. That is, if a matrix A acts on a vector v , then the product is Av and not $v^\top A$.

Definition 1.1. A map $f : \mathbb{C}^n \rightarrow \mathbb{C}$ is said to be a **linear map** over $\mathbb{C}$ if

$$f(\alpha x + \beta y) = \alpha f(x) + \beta f(y)$$

for every $x, y \in \mathbb{C}^n$ and $\alpha, \beta \in \mathbb{C}$.

Linear maps over $\mathbb{C}$ can be structured nicely as follows.

Proposition 1.2. Let $f : \mathbb{C}^n \rightarrow \mathbb{C}$ be a linear map over $\mathbb{C}$. Then there exists some $\alpha_1, \dots, \alpha_n \in \mathbb{C}$ such that

$$f(x) = f([x_1, \dots, x_n]^\top) = \sum_{i=1}^n \alpha_i x_i$$

Conversely, every map of the above form is linear.

Proof. The converse is trivial to verify. To prove the forward direction we write our vector x as

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix} + \cdots + x_n \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

Since the map f is linear we have

$$f(x) = x_1 f(e_1) + \cdots + x_n f(e_n)$$

where for $1 \leq i \leq n$, e_i is the vector in $\mathbb{C}^n$ with 1 in the i -th place and 0 elsewhere. These $f(e_i) \in \mathbb{C}$ are the required values of α_i , that is, we set $\alpha_i = f(e_i)$ to get the required form. ■

The above proposition allows us to completely characterise maps $f : \mathbb{C}^n \rightarrow \mathbb{C}^m$ which are $\mathbb{C}$ -linear in each coordinate. The resulting structural coefficients are essentially the information a matrix stores.

Since linear algebra through matrices is used to deal with solving systems of equations, we define the following matrix which will be central to most of our techniques for solving such systems.

Definition 1.3. Fix $m, n \geq 1$ and a field $\mathbb{F}$. A **linear system** (over $\mathbb{F}$) of m equations in n variables is written completely as $Ax = b$ where A is an $m \times n$ matrix, x is an unknown vector of length n and b is a known vector of length m . All the entries are taken from the same field $\mathbb{F}$.

We write $A^+ = [A \mid b]$ and call it the **augmented matrix** associated with the system.

Those who do not know what a field is (the formal definition of which you will soon encounter at a later point in the notes) may consider $\mathbb{F} = \mathbb{C}$ and always think of everything going on over the complex numbers. Real numbers also do form a field and one may also use the real numbers to think of fields at an early stage. Of course, real numbers lack the property of ‘being closed’ since $x^2 + 1 = 0$ has no roots over the reals but any polynomial with possibly complex coefficients always has all roots being complex numbers and is ‘closed’ in some sense.

We now describe some fundamental operations that may be performed on systems which preserve the solutions.

Proposition 1.4. *The following operations on a system of linear equations give a new system of linear equations which furnishes the same solution as the original one.*

- (i) *Interchanging two equations.*
- (ii) *Multiplying an entire equation with a non-zero scalar from $\mathbb{F}$.*
- (iii) *Multiplying an equation by a scalar from $\mathbb{F}$ and adding it to another equation.*

Proof. We prove each of these by writing out the system and showing in both directions that if we have a solution of one system, then it is also a solution of the other system.

- (i) Consider the following systems of equations:

$$\begin{array}{ccc}
 \sum_{k=1}^n a_{1,k}x_k = b_1 & & \sum_{k=1}^n a_{1,k}x_k = b_1 \\
 \vdots & & \vdots \\
 \sum_{k=1}^n a_{i,k}x_k = b_i & & \sum_{k=1}^n a_{j,k}x_k = b_j \\
 \vdots & & \vdots \\
 \sum_{k=1}^n a_{j,k}x_k = b_j & & \sum_{k=1}^n a_{i,k}x_k = b_i \\
 \vdots & & \vdots \\
 \sum_{k=1}^n a_{m,k}x_k = b_m & & \sum_{k=1}^n a_{m,k}x_k = b_m \\
 (S_1) & & (S_2)
 \end{array}$$

Suppose $(\alpha_1, \dots, \alpha_n)$ satisfies system S_1 then it is clear that the same also satisfies S_2 and conversely if a solution satisfies S_2 then it also satisfies S_1 . Thus, S_1 and S_2 have the same solution sets.

(ii) Let $c \in \mathbb{F} \setminus \{0\}$ be a non-zero scalar. Consider the following systems of equations:

$$\begin{array}{ccc}
\sum_{k=1}^n a_{1,k} x_k = b_1 & & \sum_{k=1}^n a_{1,k} x_k = b_1 \\
\vdots & & \vdots \\
\sum_{k=1}^n a_{i,k} x_k = b_i & & \sum_{k=1}^n (ca_{i,k}) x_k = cb_i \\
\vdots & & \vdots \\
\sum_{k=1}^n a_{m,k} x_k = b_m & & \sum_{k=1}^n a_{m,k} x_k = b_m \\
(S_1) & & (S_2)
\end{array}$$

Suppose $(\alpha_1, \dots, \alpha_n)$ satisfies system S_1 . Then the i -th equation holds, meaning $\sum_{k=1}^n a_{i,k} \alpha_k = b_i$. Multiplying both sides by the scalar c yields $\sum_{k=1}^n (ca_{i,k}) \alpha_k = cb_i$, which implies the solution satisfies S_2 . Conversely, suppose a solution satisfies S_2 . Since $c \neq 0$, the multiplicative inverse c^{-1} exists in $\mathbb{F}$. Multiplying the i -th equation of S_2 by c^{-1} recovers the i -th equation of S_1 . Thus, S_1 and S_2 have the same solution sets.

(iii) Let $c \in \mathbb{F}$ be a scalar. We multiply the i -th equation by c and add it to the j -th equation. Consider the systems:

$$\begin{array}{ccc}
\sum_{k=1}^n a_{1,k} x_k = b_1 & & \sum_{k=1}^n a_{1,k} x_k = b_1 \\
\vdots & & \vdots \\
\sum_{k=1}^n a_{i,k} x_k = b_i & & \sum_{k=1}^n a_{i,k} x_k = b_i \\
\vdots & & \vdots \\
\sum_{k=1}^n a_{j,k} x_k = b_j & & \sum_{k=1}^n (a_{j,k} + ca_{i,k}) x_k = b_j + cb_i \\
\vdots & & \vdots \\
\sum_{k=1}^n a_{m,k} x_k = b_m & & \sum_{k=1}^n a_{m,k} x_k = b_m \\
(S_1) & & (S_2)
\end{array}$$

Suppose $(\alpha_1, \dots, \alpha_n)$ satisfies system S_1 . Then both $\sum_{k=1}^n a_{i,k} \alpha_k = b_i$ and $\sum_{k=1}^n a_{j,k} \alpha_k = b_j$ hold. Multiplying the former by c and adding it to the latter gives $\sum_{k=1}^n a_{j,k} \alpha_k + c \sum_{k=1}^n a_{i,k} \alpha_k = b_j + cb_i$, which simplifies to $\sum_{k=1}^n (a_{j,k} + ca_{i,k}) \alpha_k = b_j + cb_i$. This is exactly the j -th equation in S_2 , so the solution satisfies S_2 . Conversely, if a solution satisfies S_2 , it satisfies both the i -th equation and the modified j -th equation. Subtracting c times the i -th equation from the j -th equation of S_2 yields $\sum_{k=1}^n (a_{j,k} + ca_{i,k}) \alpha_k - c \sum_{k=1}^n a_{i,k} \alpha_k = b_j + cb_i - cb_i$, which simplifies back to $\sum_{k=1}^n a_{j,k} \alpha_k = b_j$. Therefore, the solution also satisfies S_1 . ■

Performing operations algorithmically can be described in plain text easily. However, we need to formalise it into the language of matrices in order to be able to do something meaningful with them. To this extent we define the following matrices which we connect to the above proposition in a proposition below.

Definition 1.5. Given a field $\mathbb{F}$ and some positive integer $n \geq 1$, for $1 \leq i, j \leq n$ we define

$$\begin{aligned} P_{i,j} &= I - e_i e_i^\top - e_j e_j^\top + e_i e_j^\top + e_j e_i^\top \\ M_j(c) &= \text{Diag}[1, \dots, c, \dots, 1] \quad (c \text{ in the } j\text{-th position}) \\ E_{i,j}(c) &= I + c e_i e_j^\top \end{aligned}$$

Here e_i is the vector of length n over $\mathbb{F}$ with 1 in the i -th entry and 0 elsewhere.

One may notice that performing the three operations from Proposition 1.4 on the identity matrix I gives us the three respective matrices defined above. This is indeed the general idea. Instead of asking how new matrix would look when the operations are performed on a matrix A , we ask how it would look when performed on $IA = A$. Using some magical associativity law (think about this), we may apply the operations on I and then the new matrix is easy to describe as defined above.

Proposition 1.6. *In accordance with the above Proposition 1.4, each of the mentioned operations on the system of equations corresponds to the action of matrices on the augmented matrix as follows (assume standard parameters of m and n , that is, the system is $Ax = b$ with A being an $m \times n$ matrix over $\mathbb{F}$):*

- (i) *Interchanging the i -th and j -th equations is equivalent to left multiplying the augmented system by the $m \times m$ matrix $P_{i,j}$.*
- (ii) *Multiplying the j -th equation by some non-zero scalar c is equivalent to left multiplying the augmented system by $M_j(c)$.*
- (iii) *Multiplying the j -th equation by a scalar c and adding it to the i -th equation is equivalent to left multiplying the augmented matrix by $E_{i,j}(c)$.*

Proof. Suppose the augmented matrix is some $m \times q$ matrix B (where $q = n + 1$), which we can represent in terms of its rows as:

$$B = \begin{bmatrix} b_{1,1} & b_{1,2} & \cdots & b_{1,q} \\ b_{2,1} & b_{2,2} & \cdots & b_{2,q} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m,1} & b_{m,2} & \cdots & b_{m,q} \end{bmatrix} = \begin{bmatrix} R_1 \\ R_2 \\ \vdots \\ R_m \end{bmatrix}$$

where each $R_k = [b_{k,1}, b_{k,2}, \dots, b_{k,q}]$ is a row vector of length q . Observe that left-multiplying B by the vector $e_k^\top$ extracts the k -th row of B , meaning $e_k^\top B = R_k$. Furthermore, multiplying a column vector e_k by the row vector R_l yields an $m \times q$ matrix with R_l in the k -th row and zeros everywhere else. We now evaluate the action of each matrix:

- (i) For the permutation matrix $P_{i,j}$, we have:

$$\begin{aligned} P_{i,j}B &= (I - e_i e_i^\top - e_j e_j^\top + e_i e_j^\top + e_j e_i^\top)B \\ &= B - e_i(e_i^\top B) - e_j(e_j^\top B) + e_i(e_j^\top B) + e_j(e_i^\top B) \\ &= B - e_i R_i - e_j R_j + e_i R_j + e_j R_i \end{aligned}$$

Subtracting $e_i R_i$ and $e_j R_j$ from B replaces the i -th and j -th rows with zeros. Adding $e_i R_j$ and $e_j R_i$ subsequently places the old j -th row into the i -th position, and the old i -th row into the j -th position. This exactly interchanges the two equations.

- (ii) For the scaling matrix $M_j(c)$, we can rewrite it as $M_j(c) = I + (c - 1)e_j e_j^\top$. Thus:

$$\begin{aligned} M_j(c)B &= (I + (c - 1)e_j e_j^\top)B \\ &= B + (c - 1)e_j(e_j^\top B) \\ &= B + (c - 1)e_j R_j \end{aligned}$$

This operation adds $(c - 1)$ times the j -th row to the existing j -th row in B . The new j -th row becomes $R_j + (c - 1)R_j = cR_j$, leaving all other rows unaffected. This matches multiplying the j -th equation by the non-zero scalar c .

(iii) For the matrix $E_{i,j}(c)$, we have:

$$\begin{aligned} E_{i,j}(c)B &= (I + ce_i e_j^\top)B \\ &= IB + ce_i(e_j^\top B) \\ &= B + ce_i R_j \end{aligned}$$

The matrix $ce_i R_j$ consists of cR_j in its i -th row and zeros elsewhere. Adding this to B updates the i -th row from R_i to $R_i + cR_j$, which is precisely the operation of multiplying the j -th equation by c and adding it to the i -th equation. ■

Proposition 1.7. *The operation of interchanging two equations can be achieved by a sequence of the other two operations.*

Proof. The following is a one line proof of the above proposition.

$$P_{i,j} = E_{i,j}(1)E_{j,i}(-1)M_i(-1)E_{i,j}(-1)$$

One is encouraged to verify the above directly in terms of the equations themselves. ■

We now define the notion of a system being in row echelon form, which is a very convenient form to have for solving systems of equations.

Definition 1.8. A matrix is said to be in **row echelon form** if the following conditions hold:

- (i) If there is a row that consists entirely of zeros, then it is at the bottom of the system.
- (ii) For every non-zero row, the first non-zero entry (called the leading coefficient) is strictly to the right of the leading coefficient of the row above it.

The first non-zero entry in a non-zero row is called a **pivot**. The column in which a pivot appears is called a **pivot column**. The other columns are called **free columns**.

We define a stricter form of row echelon form, called reduced row echelon form, which is even more convenient to work with, as we shall see.

Definition 1.9. A matrix is said to be in **reduced row echelon form** if it is in row echelon form and the following additional conditions hold:

- (i) Every pivot is 1.
- (ii) Each pivot is the only non-zero entry in its column.

Theorem 1.10. *The row reduced echelon form of a matrix is unique.*

Proof. The proof is by induction on the number of columns. Suppose A has a single column. Then any REF is necessarily of the form

$$\begin{bmatrix} \alpha \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

by the definition of REF. If $\alpha = 0$, then the RREF is the same as the REF. If $\alpha \neq 0$, then the RREF is

$$\begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Now suppose that the result holds for any $m \times n$ matrix. We prove it for matrices of size $m \times (n + 1)$. Let A be an $m \times (n + 1)$ matrix and let B and C be two RREFs of A .

Let A' be the $m \times n$ matrix obtained by deleting the last column of A , and let B' and C' be the $m \times n$ matrices obtained by deleting the last columns of B and C , respectively. Since B and C are row equivalent to A , there exist invertible matrices E and F (each a product of elementary row matrices) such that $EA = B$ and $FA = C$. This fact is later proved in Proposition 1.14.

By the definition of matrix multiplication by columns, the j -th column of B is simply E applied to the j -th column of A . Therefore, truncating the last column yields $EA' = B'$ and $FA' = C'$. This establishes that B' and C' are row equivalent to A' .

Furthermore, since B and C are in RREF, removing their final columns cannot violate any of the structural conditions for being in RREF (the pivots remain 1, the columns with pivots remain standard basis vectors, and the staircase structure is strictly preserved). Thus, B' and C' are RREFs of A' . By the induction hypothesis, we have $B' = C'$.

It remains to show that the last columns of B and C , say b and c , are also equal. Consider the matrix $B - C$. This matrix is zero everywhere except possibly in the last column which is $b - c$. Suppose this is non-zero.

Let $z \in \mathbb{C}^{n+1}$ be such that $Az = 0$. Applying ERM's does not change the solution set of the homogeneous system, so we have $Bz = 0$ and $Cz = 0$. This implies that $(B - C)z = 0$. Writing it out we get

$$\begin{bmatrix} 0 & 0 & \cdots & 0 & b_1 - c_1 \\ 0 & 0 & \cdots & 0 & b_2 - c_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & b_m - c_m \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_{n+1} \end{bmatrix} = \begin{bmatrix} (b_1 - c_1)z_{n+1} \\ (b_2 - c_2)z_{n+1} \\ \vdots \\ (b_m - c_m)z_{n+1} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Since some $b_j - c_j$ ($1 \leq j \leq m$) is non-zero, we have $z_{n+1} = 0$. This means that any vector in the null space of A has a zero in its last coordinate. Consequently, the $(n + 1)$ -th column of A cannot be expressed as a linear combination of the first n columns (think why).

Because elementary row operations preserve linear relations among columns, the last columns of B and C (which are b and c) cannot be linear combinations of the columns of B' and C' , respectively. By the definition of RREF, a column is not a linear combination of previous columns if and only if it introduces a new pivot.

Since $B' = C'$ by the induction hypothesis, they have the exact same number of pivots, say r . Therefore, the new pivot in both B and C must be the $(r + 1)$ -th standard basis vector. Hence $b = c = e_{r+1}$, which contradicts the assumption that $b - c$ is non-zero. Therefore, we must have $b = c$, which completes the inductive step. ■

Definition 1.11. A variable x_i involved in $[A \mid b]$ is called a **simple variable** if the i -th column of the row reduced matrix of A is a pivot column. Otherwise, it is called a **free variable**. Here, A is of size $m \times n$ and b is of length m . x is the unknown vector of length n .

Proposition 1.12. All simple variables x_{a_r} associated with a solvable system $[A \mid b]$ are expressible strictly in terms of constants and free variables x_j with $j > a_r$. Consequently, the free variables are the only variables that can be assigned arbitrary values in $\mathbb{C}$ to obtain a solution to the system.

Proof. Let $a_1 < a_2 < \cdots < a_k$ be the labels of the pivot columns of A_{RREF} . We wish to solve the augmented system $[A_{\text{RREF}} \mid c]$, where the row reduction $A \xrightarrow{\text{RREF}} A_{\text{RREF}}$ induces the transformation of the vector $b \xrightarrow{\text{RREF}} c$.

Clearly, the number of pivots cannot exceed the dimensions of the matrix, so $k \leq \min\{m, n\}$. In particular, $k \leq m$, where m is the number of rows. By the definition of row echelon form, the first k rows contain the pivots, and all rows below the k -th row must necessarily be zero rows. Since the system is solvable we must have $c_{k+1} = c_{k+2} = \cdots = c_m = 0$.

Let us describe the rows of $[A_{\text{RREF}} \mid c]$ in more detail. The first row is of the form $0 \cdots 0, 1, \alpha_{i_1}, \alpha_{i_2} \cdots \alpha_{i_{l_1}}, c_1$.

The second row has the form $0 \cdots 0, 1, \beta_{i_1}, \beta_{i_2} \cdots \beta_{i_{l_2}}, c_2$. And so on until the last column which has the form $0 \cdots 0, 1, \gamma_{i_1}, \gamma_{i_2} \cdots \gamma_{i_{l_k}}, c_k$. Here, the 1 in the i -th row is located at column a_i and $l_i = n - a_i$ for $1 \leq i \leq k$.

Now, consider the equation corresponding to the r -th row for any $1 \leq r \leq k$. The leading non-zero entry is a pivot of 1 located at column a_r . By the second condition of the definition of reduced row echelon form, this pivot is the only non-zero entry in its column. Furthermore, for any other pivot column a_s (where $s \neq r$), the entry in the r -th row is forced to be 0.

Consequently, when we look at the r -th row, the coefficients corresponding to all other simple variables x_{a_s} are zeroes (they either lie above a pivot 1 or below it). Because the staircase structure of a row echelon form dictates that all entries to the left of the pivot column a_r must be zero, the only variables that can appear with non-zero coefficients in this row equation must have column indices j strictly greater than a_r and must be free variables.

Letting $u_{r,j}$ denote the scalar entry of A_{RREF} in the r -th row and j -th column, we can write out this row equation explicitly as:

$$1 \cdot x_{a_r} + \sum_{\substack{j > a_r \\ j \text{ is free}}} u_{r,j} x_j = c_r$$

Rearranging this equation allows us to isolate the simple variable x_{a_r} :

$$x_{a_r} = c_r - \sum_{\substack{j > a_r \\ j \text{ is free}}} u_{r,j} x_j$$

This explicit formulation holds independently for each row $r \in \{1, \dots, k\}$. It demonstrates that the value of every simple variable is rigidly determined by the system's constants and the chosen values of the free variables. Consequently, one cannot assign arbitrary values to the simple variables; their values are entirely dependent on the choices made for the free variables.

Conversely, the free variables do not appear as leading variables in any equation of the reduced system, meaning they are entirely unconstrained. We may therefore assign them any arbitrary values in $\mathbb{C}$, and the above equations will uniquely dictate the necessary values for the simple variables to form a valid solution vector. This establishes that the free variables are indeed the only variables that afford such arbitrary assignment. ■

We now make some statements on the relationship between the number of pivots and the number of solutions of the system.

Proposition 1.13. *Let A be in row reduced echelon form. If every row has a pivot then the system $[A \mid b]$ has at least one solution. If every column has a pivot then the system $[A \mid b]$ has at most one solution. If every row and every column has a pivot then the system $[A \mid b]$ has exactly one solution, A is square and in fact $A = I$.*

Proof. Let A be of size $m \times n$. If every row has a pivot, then $m \leq n$. If every column has a pivot, then $n \leq m$. Thus, if every row and every column has a pivot, we must have $m = n$. Since A is in RREF and has pivots in every row and every column, the only possibility is that A is the identity matrix I . In this case, the system $[I \mid b]$ has exactly one solution, which is $x = b$.

Now suppose every row has a pivot. Then $m \leq n$. This means that there are m simple variables and $n - m$ free variables. Since the system is consistent, we can assign arbitrary values to the free variables and solve for the simple variables using the equations corresponding to the pivot rows. This guarantees at least one solution.

Now suppose every column has a pivot. Then $n \leq m$. This means that there are n simple variables and no free variables. The values of the simple variables are completely determined by the equations corresponding to the pivot rows, and there are no free variables to assign arbitrary values to. Therefore, there can be at

most one solution. In fact, since every column in a pivot column, we must have

$$A = \begin{bmatrix} I_{n \times n} \\ 0_{(m-n) \times n} \end{bmatrix}$$

■

We now prove that the RREF can be computed and obtained only using elementary row operations.

Proposition 1.14. *For any matrix A , there exists a finite sequence of elementary row operations that transforms A into its row reduced echelon form.*

Proof. We describe an algorithm to compute the RREF of A using elementary row operations. Let A be an $m \times n$ matrix. We start with the first column and look for the first non-zero entry in this column. If there is no non-zero entry, we move on to the second column and repeat this process until we find a column with a non-zero entry or we exhaust all columns.

Suppose we find a non-zero entry in the i -th row and j -th column. We can use the first operation (interchanging two equations) to swap the i -th row with the first row, bringing the non-zero entry to the top of the current submatrix. Next, we can use the second operation (scaling an equation) to multiply the first row by the inverse of this non-zero entry, making it a pivot of 1. Finally, we can use the third operation (adding a multiple of one equation to another) to eliminate all other entries in the j -th column below this pivot by adding appropriate multiples of the first row to those rows.

After processing the first pivot column, we move on to the next column and repeat this process for the submatrix that excludes the rows above the current pivot row. We continue this process until we have processed all columns or all remaining rows are zero.

Once we have obtained a row echelon form, we can then perform back substitution using elementary row operations to ensure that each pivot is the only non-zero entry in its column, thus achieving reduced row echelon form.

Since each step involves a finite number of elementary row operations, and there are at most $\min(m, n)$ pivots to process, this algorithm will terminate after a finite number of steps, yielding the RREF of A . ■

Note that the above proof simply gives us the existence of a sequence of elementary row operations that transforms A into its RREF. There is no restriction on the RREF being unique. This avoids circularity since the uniqueness of RREF which was proved in Theorem 1.10 relied on the above proposition. However, the algorithm described in the above proof will always yield the unique RREF of A regardless of the choices made during the process, which is a consequence of the uniqueness of RREF.

The following is an immediate consequence of the above proposition.

Corollary 1.15. *Given a square matrix A , there exists a finite sequence of elementary row operations such that the resulting matrix is either the identity matrix I or has last row consisting of all zeroes.*

Proof. Let A be an $n \times n$ matrix. By Proposition 1.14, we can apply a finite sequence of elementary row operations to transform A into its RREF, say B . Since B is in RREF, it must have a staircase structure with pivots that are either 1 or absent. If B has pivots in every row and every column, then B must be the identity matrix I . If B does not have pivots in every row, then there must be at least one row that consists entirely of zeroes, and by the definition of row echelon form, this zero row must be at the bottom of the matrix. Thus, the resulting matrix is either the identity matrix or has its last row consisting of all zeroes. ■

We now visualise the RREF in a nice manner to understand the structure of the solution set of a system of equations. Here we need to allow the swapping of columns. Suppose A is an $m \times n$ matrix and has k pivots in its RREF. Then the first k rows of the RREF of A contain the pivots. And each pivot column has a single

1 in that pivot position. Thus, when we swap columns to bring the pivots together, the matrix takes the following form:

$$\begin{bmatrix} I_{r \times r} & F_{r \times p} \\ 0_{q \times r} & 0_{q \times p} \end{bmatrix}$$

Here, $p = n - r$ and $q = m - r$. The first r columns are the pivot columns and the last p columns are the free columns. The first r rows contain the pivots and the last q rows are zero rows. The matrix F contains the coefficients of the free variables in the equations corresponding to the pivot rows.

Motivated by the above corollary we make the following definition.

Definition 1.16. A square matrix A is said to be invertible if it can be transformed into the identity matrix I by a finite sequence of elementary row operations. If no such sequence exists, then A is said to be non-invertible or singular. If $E_1 \cdots E_m A = I$ then we call $E_1 \cdots E_m$ an inverse of A and denote it by A^{-1} .

Since matrices form a ring under addition and multiplication, the inverse is unique if it exists.

We now show that the elementary row matrices defined in Definition 1.5 are indeed invertible, which is a crucial fact that allows us to use them to perform row operations without worrying about losing invertibility. This is important because we want to be able to reverse the row operations if needed, and also because the invertibility of these matrices ensures that they do not change the solution set of the system of equations when applied to the augmented matrix.

Proposition 1.17. *All the three elementary row matrices are invertible.*

Proof. Let E be an elementary row matrix. We need to show that there exists a matrix E^{-1} such that $EE^{-1} = E^{-1}E = I$. We simply provide the inverses. It is left to the reader to verify that these are indeed inverses by direct matrix multiplication.

- (i) The inverse of the permutation matrix $P_{i,j}$ is itself, since swapping the same two rows twice returns the original matrix. Thus, $P_{i,j}^{-1} = P_{i,j}$.
- (ii) The inverse of the scaling matrix $M_j(c)$ is $M_j(c^{-1})$, since multiplying the j -th row by c and then by c^{-1} returns the original row. Thus, $M_j(c)^{-1} = M_j(c^{-1})$.
- (iii) The inverse of the matrix $E_{i,j}(c)$ is $E_{i,j}(-c)$, since adding c times the j -th row to the i -th row and then adding $-c$ times the j -th row to the i -th row returns the original row. Thus, $E_{i,j}(c)^{-1} = E_{i,j}(-c)$. ■

Note that merely having the ERM's being invertible was not that important. What is important is that they are invertible by other ERM's, that is, the inverse of an ERM is also an ERM.

We end the first introductory section with a few equivalent conditions of invertibility.

Theorem 1.18. *Let A be an $n \times n$ matrix. The following are equivalent:*

- (i) A is invertible.
- (ii) The system $[A \mid b]$ has exactly one solution for every choice of b .
- (iii) A has n pivots in any REF.
- (iv) The RREF of A is the identity matrix I .
- (v) A is a product of elementary row matrices.

Proof. (1) $\implies$ (2): Suppose A is invertible. Let b be an arbitrary vector in $\mathbb{C}^n$. Consider the system $[A \mid b]$, which corresponds to the matrix equation $Ax = b$. Left-multiplying both sides by the inverse A^{-1} , we obtain $x = A^{-1}b$. This demonstrates that $x = A^{-1}b$ is a solution, so a solution exists. To show uniqueness, suppose y is another solution such that $Ay = b$. Left-multiplying by A^{-1} yields $y = A^{-1}b$. Thus, the solution is unique.

(2) $\implies$ (3): Suppose the system $[A \mid b]$ has exactly one solution for every choice of $b \in \mathbb{C}^n$. In particular, choosing $b = 0$, the homogeneous system $Ax = 0$ has exactly one solution. Since the trivial solution $x = 0$ always satisfies the homogeneous system, it must be the unique solution. A unique solution implies that there are no free variables in the system. Because the system has n total variables and no free variables, all n variables must be simple variables. By definition, every simple variable corresponds to a pivot column, meaning A must have exactly n pivots in any REF.

(3) $\implies$ (4): Follows from Proposition 1.13.

(4) $\implies$ (5): Suppose the RREF of A is the identity matrix I . Since the RREF is obtained by applying a sequence of elementary row operations, there exists a finite sequence of elementary row matrices $E_1, E_2, \dots, E_k$ such that $(E_k \cdots E_2 E_1)A = I$. Let $E = E_k \cdots E_2 E_1$. Then E is an invertible matrix and $EA = I$, which implies $A = E^{-1}$. Using the property of inverses of products, we have $A = (E_k \cdots E_2 E_1)^{-1} = E_1^{-1} E_2^{-1} \cdots E_k^{-1}$. Since the inverse of any elementary row matrix is itself an elementary row matrix, A is a product of elementary row matrices.

(5) $\implies$ (1): Suppose A is a product of elementary row matrices, say $A = E_1 E_2 \cdots E_m$. By definition, every elementary row matrix is invertible. Because the product of any finite number of invertible matrices is also invertible, it follows that A is invertible. ■